

Design Before You Build — Paper D

HARD • 45 MINUTES • 30 MARKS

NAME

DATE

SCORE / 30

- Answer **every** question in the space provided. Show your working: method marks are awarded even when the final answer is wrong.
- No calculator and no computer. Every number on this paper works by hand.
- Where a question says **sketch**, draw plain boxes and labels only. Colour, real pictures and decoration earn nothing on a wireframe.
- Write **px** on every screen measurement. Give a contrast ratio as a number.
- The mark for each question is shown in square brackets on the right.
- No formula and no pass mark is given on this paper. If you cannot remember them, write down what you do know and work from there. Relative luminance L is printed wherever you need it.
- Several questions describe products and situations you may not have seen before. Apply the ideas you already have.

SECTION 1 – STRUCTURE AND PROCESS

- 1.** The Railway Club website has three pages: Home, About and Contact. The navigation on every page links to every other page. The club now adds a fourth page, **Gallery**.

- (a)** In the box, sketch a sitemap for the four-page site, with Home at the top. Show [2]
how the pages connect.

SITEMAP

- (b)** Calculate the total number of navigation links between pages before Gallery [2]
was added, and after.

2. Three entries from a design team's diary.

- P** Week 2. The team groups fifty interview notes into four themes and finds one pain that keeps repeating.
- Q** Week 4. The team draws three different layouts and tests each with five users.
- R** Week 7. Testing shows the chosen design works well, but users say it solves a problem they never had: the team had misread its own notes. They rewrite the problem statement.

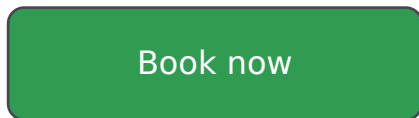
(a) Name the phase of the double diamond that entry **P** belongs to. [1]

(b) Name the phase that entry **Q** belongs to. [1]

(c) In entry **R** the team goes back. Name the phase they return to. [1]

3. A booking app has a white label on a green button. The green has relative luminance $L = 0.25$; white has $L = 1.00$.

16 px regular



text #FFFFFF L = 1.00
back #309C51 L = 0.25

20 px bold



text #FFFFFF L = 1.00
back #309C51 L = 0.25

24 px regular

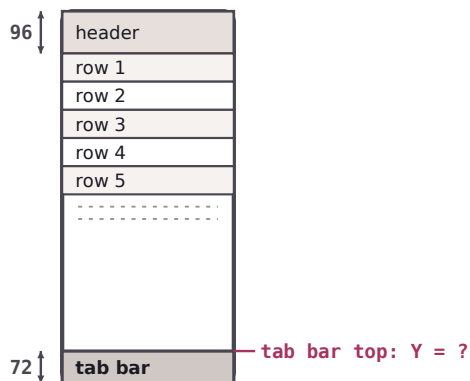


text #FFFFFF L = 1.00
back #309C51 L = 0.25

(a) Calculate the contrast ratio of the label. [1]

(b) The designer tries three labels: **16 px regular**, **20 px bold** and **24 px regular**. State which of them pass, and why. [2]

4. A frame is **390 px wide and 844 px high**. A header occupies $Y = 0$ to $Y = 96$. A bottom tab bar is **72 px** high and sits flush with the bottom edge. Between them is a list of rows, each **64 px** high with no gaps, starting at **$Y = 96$** .



FIGMA PANEL

Frame: W 390 H 844
Header: $Y = 0$ to $Y = 96$
Tab bar: H 72, flush with the bottom edge of the frame
List rows: H 64 each, no gaps, row 1 starts at $Y = 96$
Rows fill the space between the header and the tab bar.

- (a) Calculate the Y position of the top edge of the tab bar. [1]

- (b) Calculate how many whole rows fit between the header and the tab bar, and how many px are left over. [2]

- (c) Each row is a component instance. The designer edits the main component so each row is **72 px** high. State how many whole rows now fit, and which instances change. [2]

5. Diya orders lunch on her family's shared tablet, and the canteen app logs her out every time. Her journey is scored from 1 (miserable) to 5 (delighted). Two teams each make one change. Team A polishes the screens. Team B changes how signing in works.

	1 Welcome	2 Choose food	3 Sign in	4 Pay	5 Order number
Before	4	3	1	3	5
Fix A	5	5	1	4	5
Fix B	4	3	3	3	5

(a) Calculate the average score before, after Fix A and after Fix B. [2]

(b) State the lowest point before either fix, and whether Fix A changes it. [1]

(c) Fix A has the higher average. Explain why Fix B is still the better fix. [2]

SECTION 3 – HANDOVER AND DIAGNOSIS

6. Farhan has finished the Railway Club wireframes. He emails the three images to his friend Kabir, who will build the site, and sends nothing else. List **four** things Kabir cannot guess from the images alone and would have to ask about. [4]

7. Another student, Rohan, has answered two questions on this topic. Both answers are wrong. For each one, state the mistake and write out a correct version, giving a reason or example that would convince him.

	ROHAN'S ANSWERS
	(i) <i>"UX is how the app looks and UI is how it works. So changing the Pay button from grey to green is a UX fix."</i>
	(ii) <i>"Our form shows a mistake by turning the box red, so every user can see which box to fix."</i>

(i) Diagnose and correct the first answer.

[3]

(ii) Diagnose and correct the second answer.

[3]

Each answer has more than one thing wrong with it. Deal with them separately.

End of paper. Check that every screen measurement carries px, and that every UI or UX answer says why, not just which.